

Relational Algebra (RDMS)

SQL ——— RA ——— Computer
(logical plan) (Physical Plan)

Notation:

① Project / Column (π)

② Tuple / Row (σ)

③ Rename (ρ)

④ Assign ($\leftarrow$)

FROM
↓
WHERE
↓
GROUP BY
↓
HAVING
↓
SELECT
↓
ORDER BY

☐ SELECT id FROM student WHERE marks > 80

$\pi_{id}(\sigma_{marks > 80}(student))$ | $R \leftarrow \sigma_{marks > 80}(student)$
 $\pi_{id}(R)$

id	marks
A	90
B	81
C	79

☐ SELECT * FROM student WHERE Dept = 'CSE' AND att > 10

$\pi_{id, name, Dept, att}(\sigma_{Dept = 'CSE' \text{ AND } att > 10}(student))$
OR
 $\sigma_{Dept = 'CSE' \text{ AND } att > 10}(student)$

id	name	Dept	att
		CSE	16
		CS	
		CSE	12

student

☐ Rename id to sid

$\rho_{(sid, name)}(student)$

id	Name

student

☐ Change table name to s

$\rho_s(sid, name)(student)$

☐ Find $\frac{(a+b/2)}{n} / \frac{(b*0.5)}{d}$

$$n = a + b/2;$$

$$d = b * 0.5;$$

$$result = n/d;$$

☐ Find id of students who got more than 80

$R \leftarrow \sigma_{m > 80}(student)$

$\pi_{id}(R)$

Union



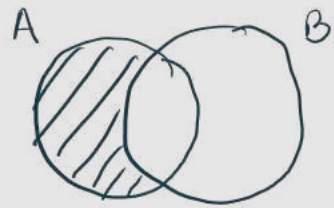
$A \cup B$

A or B



$A \cap B$

A and B



$A - B$

Find people who are not in B.

student

<u>id</u>	name
1	A
2	B
3	C

Grade

<u>id</u>	Ccode	Grade
1	a	A
2	b	A+
3	c	A

Cartesian Product

একতর ভগ্নী Table

student $\bowtie$ Grade
student.id = Grade.id

RA Tree

$\pi_{id} (\sigma_{grade > 80} (student))$
↑ Leaf

π_{id}
|
 $\sigma_{grade > 80}$
↓
student

Write RA to convert the following SQL queries.

15. SELECT distinct SName, dept FROM Student WHERE dept = 'EEE';
16. SELECT * FROM StudentCourse WHERE Grade = 'A';
17. SELECT distinct FIN, FName FROM Faculty, FacultyCourses, Course WHERE Faculty.FIN = FacultyCourses.FIN AND FacultyCourses.CCode = Course.CCode AND Course.CName = 'Automata and Computability';
18. SELECT distinct SID, SName, CName FROM Student, ST, Course WHERE Student.SID = ST.SID AND ST.assignedCourse = Course.CCode;

(15)

$\pi_{SName, dept} (\sigma_{dept='EEE'} (student))$

(17)

$\pi_{FIN, FName} (\sigma_{CName='AC'} (Faculty \bowtie_{FIN=FIN} FacultyCourse) \bowtie_{Ccode=Ccode} Course)$

(18)

$\pi_{SID, SName, CName} ((student \bowtie_{SID=SID} ST) \bowtie_{AC=Ccode} Course)$

Example Problems - I

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)

ST(SID, assignedCourse)

Faculty(FIN, FName, LName, Desk#)

Course(CCode, CName)

FacultyCourses(FIN, CCode, Section)

-- FIN references Faculty(FIN), CCode references Course(CCode)

A) Write RA for the following:

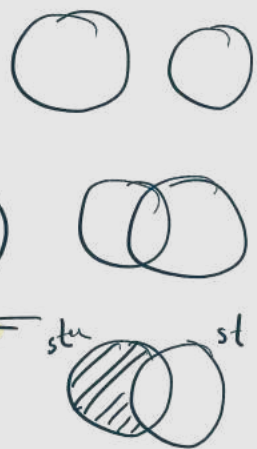
1. Find students with at least 70 credits completed from 'CS' dept.
2. Find FName and Desk# of all faculties.
3. Find SID, SName of Students from 'CSE' dept.

4. Find the names of students who are not an ST.

$$(1) \quad \sigma_{c \geq 70} (\text{student}) \\ \text{AND Dept} = \text{'CS'}$$

$$(2) \quad \pi_{\text{FName}, \text{Desk\#}} (\text{Faculty})$$

$$(3) \quad \pi_{\text{SID}, \text{SName}} \left(\sigma_{D = \text{'Cse'}} (\text{student}) \right)$$

$$(4) \quad \pi_{\text{name}} (\text{student}) - \pi_{\text{name}} (\text{student} \bowtie_{\text{SID} = \text{SID}} \text{ST})$$


Example Problems - II

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)

ST(SID, assignedCourse) -- SID references Student(SID)

→ Faculty(FIN, FName, Desk#)

→ Course(CCode, CName)

→ FacultyCourses(FIN, CCode, Section)

-- FIN references Faculty(FIN), CCode references Course(CCode)

B) Write RA for the following:

1. Rename CCode, CName columns of Course table to Code, CTitle
2. Find ST's SName, dept, assignedCourse.
3. Find Faculties FName and Desk# for those who are taking either CSE370 or CSE230 Courses.

4. Find Faculties FName, Desk# for those who are taking either 'Database Systems' or 'AI' Courses.

$$(1) \quad \rho_{(\text{Code}, \text{CTitle})} (\text{Course})$$

$$(2) \quad \pi_{\text{SName}, \text{dept}, \text{AC}} \left(\text{ST} \bowtie_{\text{sid} = \text{sid}} \text{student} \right)$$

(3)

$$R \leftarrow \left(\sigma_{\substack{AC = 'CSE370' \\ \text{OR } AC = 'CSE230'}} \left(\text{Faculty} \bowtie_{\text{in=in}} \text{Faculty Courses} \right) \right)$$

$$\pi_{\substack{\text{Fname}, \\ \text{Desk \#}}} (R)$$

(d)

$$\pi_{\substack{\text{Fname}, \\ \text{Desk \#}}} \left(\left(\text{Faculty} \bowtie_{\text{FIN=FIN}} \text{Faculty Courses} \right) \bowtie_{\text{Ccode=Ccode}} \text{Course} \right)$$